

XB-10 User Guide

Version 1.1

Release Date June. 28th, 2022

Revision History

Version	Release Date	Content
1.1	06/28/2022	Remove the description of U25(Si570) for it's no longer used and will be replaced by a new component in the future version.
1.0	02/23/2022	Initial release.

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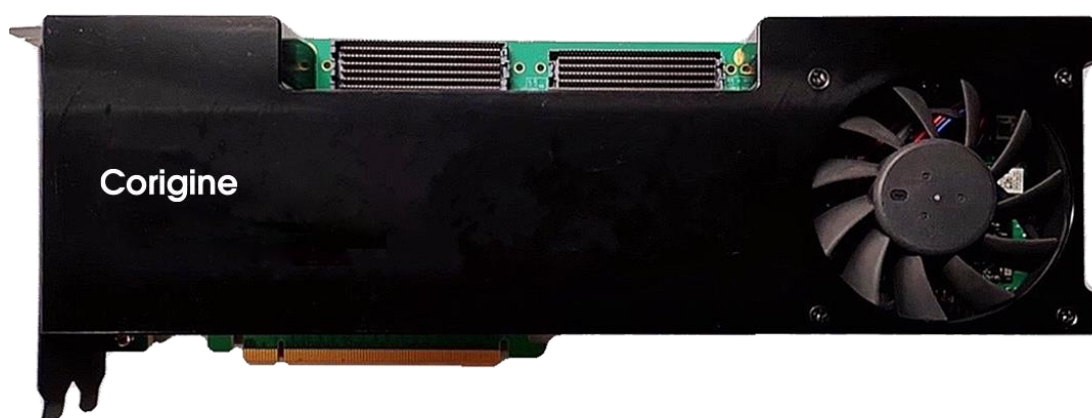
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1 Introduction

1.1 Overview

The XB-10 evaluation board for the Xilinx® Virtex® UltraScale+™ FPGA provides a hardware environment for developing and evaluation designs targeting the UltraScale+™ XCVU19P-FSVB3824 device.

Figure 1 XB-10



The XB-10 evaluation board provides following features common to many evaluation systems.

- DDR4 component memory
- Quad Small Form-factor Pluggable Double Density (QSFP-DD) connector
- 16-lane PCI Express® Interface
- Ethernet PHY
- General purpose I/O
- URAT interface
- PMOD

More features can be supported by using modules compatible with the VITA-57.1 FPGA mezzanine card (FMC) and VITA-57.4 FPGA mezzanine card plus high serial pin (FMC+HSPC) connectors on the XB-10 board.

1.2 Features

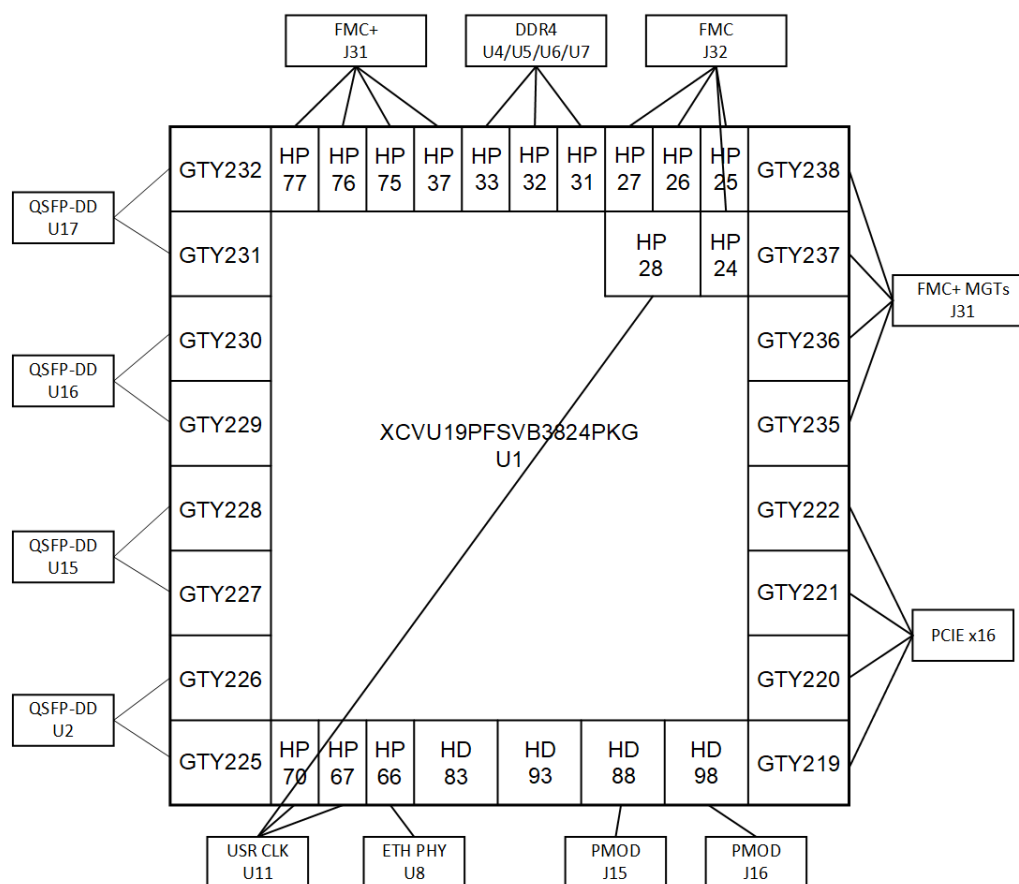
The XB-10 evaluation board features are listed here. Details for each feature are described in [Component Description](#).

- Virtex® UltraScale+™ XCVU19P-FSVB3824
- Arm® SimpleLink™ MSP432P401R microcontrollers (MCUs)

- Four 16Gb DDR4 SDRAM component memory
- 2Gb (256Mb*8) Quad SPI flash memory
- USB JTAG interface using a Digilent module with separate micro-B USB connector
- Clock sources:
 - SI5341B dec clock generator
 - Three SI570 I²C programmable LVDS clock generators
 - Ten Subminiature version A (SMA) connectors (differential)
- 64 GTY transceivers (16 Quads)
 - FMC+ HSPC connector (16 GTY transceivers)
 - 4*28 Gb/s QSFP-DD connectors (32 GTY transceivers)
 - PCIe 16-lane edge connector (16 GTY transceivers)
- PCI Express® endpoint connectivity
 - Gen1 16-lane (*16)
 - Gen2 16-lane (*16)
 - Gen3 8-lane (*8)
 - Gen3 16-lane (*16, XB-10 Rev.A V_{CCINT}=0.85V)
- Ethernet PHY SGMII interface with RJ-45 connector
- USB-to-UART bridge with micro-B USB connector
- I²C bus
- Status LEDs
- User I/O
- Two PMOD 2*6 connectors
- VITA 57.4 FMC+ HSPC connector J31
- VITA 57.1 FMC HPC1 connector J32
- Reset Button
- Configuration options:
 - Quad SPI flash memory
 - Digilent USB configuration module

1.3 Block Diagram

Figure 2 XB-10 Evaluation Board Block Diagram



1.4 Specification

Dimensions

Length	Width	Height
12.185 inch/30.95 cm	4.595 inch/11.67 cm	0.127 inch/0.323 cm ($\pm 10\%$)
Note: The XB-10 board's width exceeds the standard width of a PCI Express® card, which is 4.376 inch/11.15cm.		

Operating & Storage Requirement

Temperature	- Operating: 0°C to +45°C - Storage: -25°C to +60°C
Humidity	10% to 90%, non-condensing
Voltage	+12 V _{DC}

2 Component Description

This section describes the detailed functions of XB-10's components and features.

2.1 Overview of Board Resource

XB-10 evaluation board provides several kinds of resources (ports/interface/connectors/buttons) for engineers. For details of these resources, please refer to **Table 1 Resource List**.

Table 1 Resource List

Resource	Quantity	Note
ETH Port	1	RJ-45
UART	1	Micro USB AB type
Power Connector	1	8POS 4.2MM
JTAG	1	14POS 2MM
JTAG	1	Micro USB AB type
PMOD	2	12POS 2.54MM
FMC+	1	14 Row 0.050 Pitch
FMC	1	10 Row 0.050 Pitch
QSFP-DD	1	76 Position 0.80MM Pitch
Reset Button	1	-

2.2 Virtex® UltraScale+™ XCVU19P-FSVB3824

The XB-10 board is populated with the Virtex® UltraScale+™ XCVU19P-FSVB3824. For more information on Virtex® UltraScale+™ FPGAs, see Datasheet on [Xilinx](#).

When Creating FPGA designs for the XB-10, one could use Xilinx Vivado tool to choose the appropriate V_{CCINT} , which is 0.85V for XCVU19P-FSVB3824.

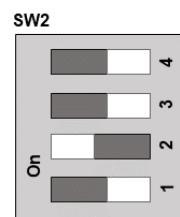
FPGA Configuration

XB-10 board supports two FPGA configuration modes:

- Quad SPI flash memory
- Digilent USB configuration module

To activate the flash configuration mode, engineers need to adjust the Switch (SW2) on XB-10 board to the following setting.

Figure 3 SW2 Setting



Configuration Mode	Switch Setting [1:4]
Quad SPI flash memory	0100

I/O Voltage Rails

There are 36 I/O banks available on the XCVU19P-FSVB3824. The voltages applied to the FPGA I/O banks used by XB-10 are listed in **Table 2 I/O Voltage Rails**.

Table 2 I/O Voltage Rails

FPGA Bank	Power Supply Rail Net Name	Voltage
Bank 24	VCC_VADJ1	1.8V
Bank 25	VCC_VADJ1	1.8V
Bank 26	VCC_VADJ1	1.8V
Bank 28	VCC_1_8V	1.8V
Bank 29	VCC_1_8V	1.8V
Bank 31	VCC_1_2V	1.2V
Bank 32	VCC_1_2V	1.2V
Bank 33	VCC_1_2V	1.2V
Bank 65	VCC_1_8V	1.8V
Bank 66	VCC_1_8V	1.8V
Bank 67	VCC_1_8V	1.8V
Bank 70	VCC_1_8V	1.8V
Bank 75	VCC_VADJ2	1.8V
Bank 76	VCC_VADJ2	1.8V
Bank 77	VCC_VADJ2	1.8V
Bank 83	VCC_3_3V	3.3V
Bank 88	VCC_3_3V	3.3V
Bank 93	VCC_3_3V	3.3V
Bank 98	VCC_3_3V	3.3V

2.3 DDR4 Component Memory

The DDR4 component memory system is comprised of four DDR4 SDRAM devices.

- Manufacturer: Micron

Component Description

- Part Number: MT40A1G16KD-062E
- Supply Voltage: 1.2V
- Density: 16Gb

The XB-10 DDR interface performance is documented in the [Virtex Ultrascale+ FPGA Data Sheet: DC and AC Switching Characteristics](#).

The connections between the DDR4 component memories and XCVU19P are listed in [Pin Assignment for DDR](#).

2.4 Quad SPI Flash Memory

The XB-10 board host a NOR flash configuration memory U14.

- Manufacturer: Micron
- Part Number: MT25QU02GCBB8E12-0SIT
- Supply Voltage: 1.7V-2.0V
- Density: 2Gb
- Datapath Width: 8 bits
- Data Rate: Up to 90 Mb/s

2.5 Clock Generation

The XB-10 evaluation board provides multiple clock sources to the FPGA. Clock sources to FPGA connections are listed in [Pin Assignment for CLK](#).

2.6 28 Gb/s QSFP-DD Connectors

The XB-10 board contains four Quad Small Form-factor Pluggable Double Density (QSFP-DD) connector. Each connector is housed within a single 28Gb/s QSFP-DD cage assembly.

2.7 10/100/1000 Mb/s Ethernet PHY

The XB-10 evaluation board uses the TI® PHY device DP83867ISRZ (U8) for Ethernet communications at 10Mb/s, 100Mb/s, or 1,000 Mb/s. The PHY connection to a user-provided Ethernet cable is through J1 (RJ-45 connector), a Wurth 7499111221A with built-in magnetics and status LEDs.

2.8 FPGA Mezzanine Card Interface

The XB-10 evaluation board supports the VITA 57.1 FPGA Mezzanine Card (FMC) specification by providing a subset implementation of the High Pin Count (HPC) connector at J32. HPC connectors are populated with 400 pins in a 10*40 form.

In addition, the XB-10 evaluation board supports the VITA 57.4 FPGA Mezzanine Card Plus (FMC+ or FMCP) specifications by providing a subset implementation of the High Serial Pin Connector (HSPC) at J31. HSPC connectors are populated with 560 pins in a 14*40 form. The connector is keyed so that a mezzanine card, when installed on the XB-10 evaluation board, faces away from the board.

J32 FMC Connector Type

Samtec® SEAF™ series, 1.27 mm (0.050 inch) pitch. Mates with SEAM series connector. More information about SEAF series connectors is available on Samtec. More information about the VITA 57.1 FMC specification is available on VITA FMC Marketing Alliance.

The 400-pin HPC connector defined by the FMC specification provides connectivity for up to:

- 160 single-ended or 80 differential user-defined signals
- 10 transceiver differential pairs
- Two transceiver differential clocks
- Four differential clocks
- 159 GND and 15 PWR connections

FMC HPC Connector J32

The HPC connector at J32 implements a subset of the full FMC HPC connectivity.

- 68 single-ended or 34 differential user-defined pairs (34 LA pairs: LA[00:33])
- Two differential clocks
- 159 GND and 15 PWR connections

J31 FMC+ Connector Type

Samtec® SEAF™ series, 1.27 mm (0.050 inch) pitch. Mates with SEAM series connector. More information about SEAF series connectors is available on Samtec. More information about the VITA 57.1 FMC specification is available on VITA FMC Marketing Alliance.

The 560-pin HPC connector defined by the FMC specification provides connectivity for up to:

Component Description

- 160 single-ended or 80 differential user-defined signals
- 24 transceiver differential pairs
- Six transceiver differential clocks
- Four differential clocks
- 239 GND and 15 PWR connections

FMC+ Connector J31

The HSPC connector at J31 implements a subset of the full FMC+ connectivity.

- 116 single-ended or 58 differential user defined pairs (34 LA pairs: LA[00:33]; 24 HA pairs: HA[00:33])
- 24 transceiver differential pairs
- Six transceiver differential clocks
- Two differential clocks
- 239 GND and 16 PWR connections

2.9 USB-UART

The XB-10 board contains a Silicon Labs CP2105 USB-to-UART bridge device (U9) which allows a connection to a host computer with a USB port. The USB cable is supplied in the XB-10 Evaluation Kit (Type-A end to host computer; Type Micro-B USB end to XB-10 board connector J3). The CP2105 is powered by the USB 5V provided by the host PC when the USB cable is plugged into the USB port on the XB-10 board.

Xilinx UART IP is expected to be implemented in the FPGA logic. The FPGA supports the USB-to-UART bridge using four signal pins: Transmit (TX), Receive (RX), Request to Send (RTS), and Clear to Send (CTS).

Silicon Labs provides royalty-free Virtual COM Port (VCP) device for the host computer. These drivers permit the USB-to-UART bridge to appear as a COM port to communications application software that runs on the host computer. The VCP device drivers must be installed on the host PC prior to establishing communications with the XB-10 Board.

The XB-10 USB-to-UART provides basic commands to change board setting such as FMC/FMC+ voltage and user clock frequency.

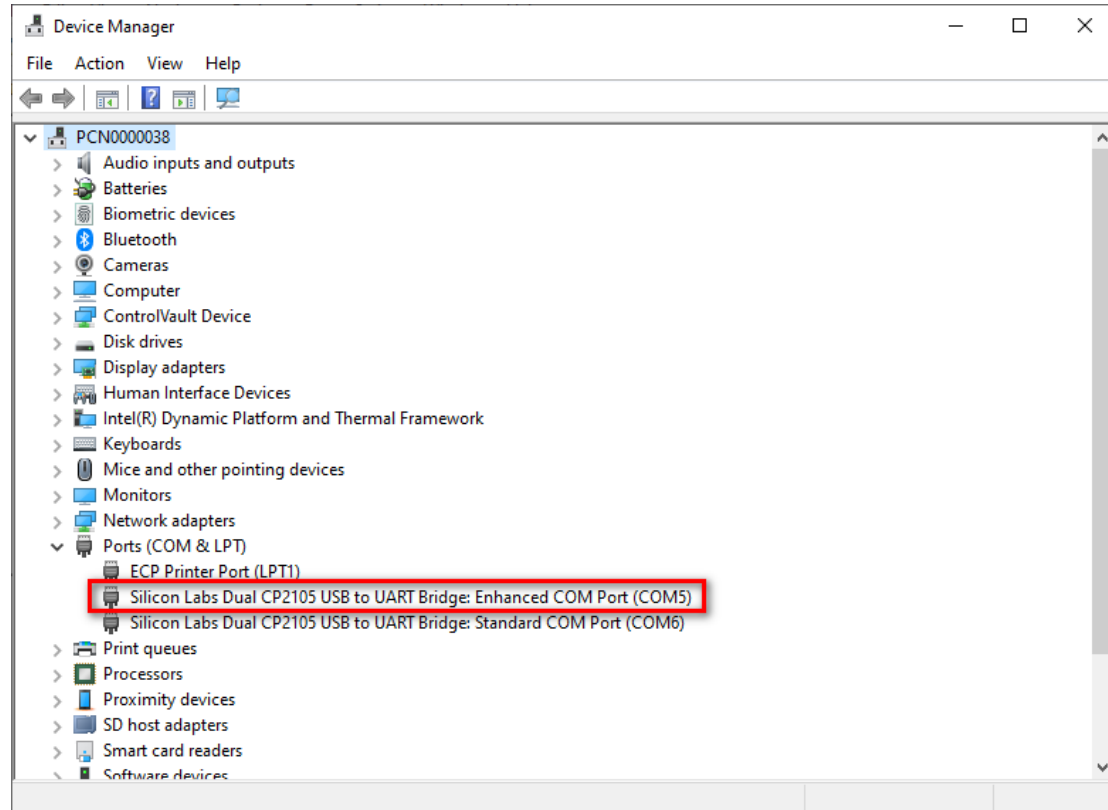
Setup UART

The XB-10 USB-to-UART provides an UART-over-USB connection to communicate with onboard MCU to execute I2C protocol. A micro-USB cable and PC terminal application such as Hyperterminal or Putty are needed to interface with COM port.

Component Description

On the host, a virtual COM port for XB-10 is generated when XB-10 is connected. You need to identify the COM port for the XB-10 MCU. On Windows PCs, Device Manager can assist.

Note: Ensure the Silicon Labs VCP USB-UART drivers are installed on the host PC. See [Silicon Labs CP210x USB-to-UART Installation Guide \(UG1033\)](#).



The UART for XB-10 MCU is “Silicon Labs Dual CP2105 USB to UART Bridge: Enhanced COM Port”. In this case, Device Manager shows COM5, but this port can vary from one host PC to another. After you identify the COM port, configure it in your host application along with following setting for serial port.

Basic commands

The basic commands only work with manufactory default settings, and the input clock source must be 100MHz.

- a. help: display available commands
- b. select_pll_clock_source: select onboard PLL clock source
 - a) SYNOPSIS
Select_pll_clock_source [0|1]
 - b) DESCRIPTION
0: external 100MHz (for synchronizing multiple XB-10)
1: internal 100MHz
 - c) EXAMPLE
select_pll_clock_source0
select_pll_clock_source1

Component Description

Note: This command will feedback current source if no clock source is selected.

- c. `set_clock_output_divider`: set user clock output divider to adjust clock frequency.
User clock output frequency = $\frac{1000}{(var+1) \times 2}$ MHz.

For example, if var=0x4, the the output frequency will be 250MHz.

- a) SYNOPSIS

`set_clock_output_divider <1|2|3|4|5>[value]`

- b) DESCRIPTION

1-5:user clock selection

Value: divider value (hex)

- c) EXAMPLE

`set_clock_output_divider 1`

`set_clock_output_diver 1 5`

Note: This command will feedback current value if no [value] is selected.

- d. `set_fmc_voltage`: set up FMC/FMC+ connector output voltage. The out-of-box voltage was set to 1.8V and turned on after powering on the board. Once the voltage is changed by this command, the power on the connector will be off after powering on board. You muse issue the command to turn it on.

- a) SYNOPSIS

`set_fmc_voltage<0|1> [ [on [1.0v|1.2v|1.35v|1.5v|1.8v] ] |off]`

- b) DESCRIPTION

0: FMC

1: FMC+

on|off: Power on/off

- c) EXAMPLE

`set_fmc_voltage 0`

`set_fmc_voltage 0 on`

`set_fmc_voltage 0 on 1.2v`

`set_fmc_voltage 0 off`

Note: When XB-10 is powering up, the daughter card on FMC/FMC+ will be detected. If a card was detected with valid VADJ record, the voltage will be set and turn on automatically, otherwise the power on FMC/FMC+ remains off to protect the board. User must use this command to set desired voltage and turn on the power.

- e. `detect_fmc`: detecting daughter card on FMC/FMC+ connector. If a was detected, report card's board information if any.

- a) SYNOPSIS

`detect_fmc<0|1>`

- b) DESCRIPTION

0: FMC

1: FMC+

- c) EXAMPLE

`detect_fmc 0`

detect_fmc 0

Advance commands



CAUTION: Advanced commands enables user to change any value of connected I²C devices on the XB-10.

- a. write: write data through I²C bus.
 - a) SYNOPSIS
write slave_addr [reg_addr] data
 - b) DESCRIPTION
slave_addr: I²C device address (hex)
reg_addr: optional register address, they will be sent out before data (hex)
data: data will be sent to I²C bus (hex)
 - c) EXAMPLE
write 50 0 B4B3B2B1
- b. read: read data through I²C bus.
 - a) SYNOPSIS
read slave_addr [reg_addr] len
 - b) DESCRIPTION
slave_addr: I²C device address (hex)
reg_addr: optional register address, they will be sent out before data (hex)
len: number of bytes to be read
 - c) EXAMPLE
read 50 1 4
- c. write_clock_reg: write clock generator's register (SI5341B)
 - a) SYNOPSIS
write_clock_reg reg_addr data
 - b) DESCRIPTION
reg_addr: 16 bits register address. (hex)
data: data to be wrote to
 - c) EXAMPLE
write_clock_reg 024A 4
- d. read_clock_reg: read clock generator's register (SI5341B)
 - a) SYNOPSIS
read_clock_reg reg_addr len
 - b) DESCRIPTION
reg_addr: 16 bits register address. (hex)
len: number of bytes to be read
 - c) EXAMPLE
read_clock_reg 024A 3

2.10 USB-JTAG

JTAG configuration is provided through a Digilent onboard USB-to-JTAG configuration logic module where a host computer accesses the XB-10 board JTAG chain through a type-A (host side) to micro-B (board side J2) USB cable.

A 2 mm JTAG header is also provided in parallel for access by Xilinx download cables, such as the Platform Cable USB II. JTAG configuration is allowed at any time regardless of FPGA mode pin settings.

When a FMC Card is attached to the XB-10 board, it is automatically added to the JTAG chain through electronically controlled single-pole single-throw (SPST) switched U27 and U28. The SPST switches are in a normally closed state and transition to an open state when FMC card is attached. Switch U27 adds an attached FMC1 HPC mezzanine card to the FPGAs JTAG chain as determined by the FMC_HPC_PRSENT_M2C_B signal. Switch U28 adds an attached FMC23 HPC mezzanine card to the FPGAs JTAG chain as determined by the FMC_LPC_PRSENT_M2C_B signal. The attached FMC card must implement a TDI-to-TDO connection via a device or bypass jumper to ensure that the JTAG chain connects to the FPGA U1.

2.11 I2C Bus

XB-10 evaluation board implants a single I²C port on the FPGA (FPGA_VU19P_SDA, FPGA_VU19P_SCL), which is routed through two 1-to-8 channel I²C bus switch (U20&U31). The bus switch can operate at speeds up to 400kHz.

2.12 Power Management

The following list demonstrates the voltage rail of every module

Reference Designator	Device Type	Power Rail Voltage
P1	BMR4614001	0.9VA
P2	BMR4613001	1.8V
P3	BMR4613001	2.5V
P4	BMR4613001	1.2VA
P5	BMR4613001	3.3V
P6	BMR4613001	1.2V
P7	BMR4613001	1.8V
P8	BMR4613001	1.8V
P9	BMR4670010	0.85V

3 Setup and Configuration

3.1 Electrostatic Discharge Caution



CAUTION: Electrostatic Discharge (ESD) can damage electronic components and lead to total or intermittent failures, when they are improperly handled. Be aware to follow ESD-prevention procedures when removing and replacing components.

ESD-Prevention Procedure

- Wear an ESD wrist or ankle strap and maintain skin contact with the strap. Connect the strap end to an unpainted metal surface on the chassis.
- Prevent the adapter from contacting with your clothing. The strap protects component from ESD generated by human body only.
- Handle the adapter by its bracket or edges only. Avoid touching the printed circuit board or the connectors.
- Always place the adapter on an antistatic surface such as the bag supplied in the kit.
- The adapter must be stored in an antistatic bag during transportation.

3.2 Installing the XB-10 Board in a PC Chassis

The XB-10 board needs to be installed inside a computer chassis to work or test PCI Express® function.

When the XB-10 board is installed inside computer chassis (plugged into the PCIe® slot), power is supplied by the ATX power supply.

Installation Procedure

1. Remove the packings of the XB-10 Board.
2. Power down the host computer and remove the power cord from the PC.
3. Open the PC chassis following the instructions from the PC manufacturer.
4. Select a vacant PCIe expansion slot and remove the expansion cover (on the back of the chassis) by removing the screws on the top and bottom of the cover.
5. The XB-10 board requires three adjacent PCIe® slots because its powerful cooling fan exceeds the PCIe® top side component height restriction. Make sure the slots closest to the front and back of the board are free of obstructions.
6. Plug the XB-10 board into the center of the three open slots.

7. Install the top mounting bracket screw into the PC expansion cover retainer bracket to secure the XB-10 board in its slot.

8. Connect the ATX power supply to the XB-10 board using the ATX power supply adapter cable»

- a) Plug the 8-pin 2*4 Molex connector on the adapter cable into J5 on the XB-10 board.
- b) Plug the peripheral power connector from the ATX power supply into the adapter cable connector.



CAUTION: Do not plug a PC ATX power connector into J5 on the XB-10 evaluation board. This may damage the XB-10 board and void the board warranty.

4 Pin Assignment

4.1 Pin Assignment for DDR

Net Name	Pin Name
DDR4_A[0]	N52
DDR4_A[1]	P52
DDR4_A[2]	V54
DDR4_A[3]	V53
DDR4_A[4]	N54
DDR4_A[5]	N53
DDR4_A[6]	U54
DDR4_A[7]	U53
DDR4_A[8]	R53
DDR4_A[9]	R52
DDR4_A[10]	T53
DDR4_A[11]	T52
DDR4_A[12]	V56
DDR4_A[13]	R55
DDR4_ACT_N	P54
DDR4_ALERT_N	V63
DDR4_BA[0]	R54
DDR4_BA[1]	U56
DDR4_BG[0]	U55
DDR4_CK_T	N56
DDR4_CK_C	N57
DDR4_CKE	T56
DDR4_RAS_N_A16	P57
DDR4_CAS_N_A15	R57
DDR4_WE_N_A14	T55
DDR4_CS_N	T60
DDR4_DM[0]	AH61
DDR4_DM[1]	AE62
DDR4_DM[2]	AB61
DDR4_DM[3]	Y62
DDR4_DM[4]	AC56
DDR4_DM[5]	AC54
DDR4_DM[6]	AB51

Pin Assignment

DDR4_DM[7]	Y47
DDR4_DQ[0]	AH58
DDR4_DQ[1]	AH59
DDR4_DQ[2]	AH63
DDR4_DQ[3]	AG63
DDR4_DQ[4]	AF60
DDR4_DQ[5]	AF61
DDR4_DQ[6]	AG58
DDR4_DQ[7]	AF58
DDR4_DQ[8]	AF63
DDR4_DQ[9]	AE63
DDR4_DQ[10]	AC58
DDR4_DQ[11]	AC59
DDR4_DQ[12]	AD59
DDR4_DQ[13]	AD60
DDR4_DQ[14]	AD61
DDR4_DQ[15]	AC61
DDR4_DQ[16]	AB60
DDR4_DQ[17]	AA60
DDR4_DQ[18]	AC62
DDR4_DQ[19]	AC63
DDR4_DQ[20]	AB63
DDR4_DQ[21]	AA63
DDR4_DQ[22]	AA58
DDR4_DQ[23]	Y58
DDR4_DQ[24]	W59
DDR4_DQ[25]	W60
DDR4_DQ[26]	Y59
DDR4_DQ[27]	Y60
DDR4_DQ[28]	W61
DDR4_DQ[29]	W62
DDR4_DQ[30]	V58
DDR4_DQ[31]	V59
DDR4_DQ[32]	AA57
DDR4_DQ[33]	Y57
DDR4_DQ[34]	AB56
DDR4_DQ[35]	AA56
DDR4_DQ[36]	AB55
DDR4_DQ[37]	AA55
DDR4_DQ[38]	Y55

Pin Assignment

DDR4_DQ[39]	W55
DDR4_DQ[40]	Y54
DDR4_DQ[41]	W54
DDR4_DQ[42]	AC53
DDR4_DQ[43]	AB53
DDR4_DQ[44]	AA53
DDR4_DQ[45]	Y53
DDR4_DQ[46]	AA52
DDR4_DQ[47]	Y52
DDR4_DQ[48]	AB50
DDR4_DQ[49]	AA50
DDR4_DQ[50]	Y49
DDR4_DQ[51]	W49
DDR4_DQ[52]	Y50
DDR4_DQ[53]	W50
DDR4_DQ[54]	AA48
DDR4_DQ[55]	Y48
DDR4_DQ[56]	AC46
DDR4_DQ[57]	AC47
DDR4_DQ[58]	W45
DDR4_DQ[59]	W46
DDR4_DQ[60]	AA45
DDR4_DQ[61]	Y45
DDR4_DQ[62]	AB46
DDR4_DQ[63]	AA46
DDR4_DQS_T[0]	AG59
DDR4_DQS_C[0]	AG60
DDR4_DQS_T[1]	AE59
DDR4_DQS_C[1]	AE60
DDR4_DQS_T[2]	AB58
DDR4_DQS_C[2]	AB59
DDR4_DQS_T[3]	V61
DDR4_DQS_C[3]	V62
DDR4_DQS_T[4]	W56
DDR4_DQS_C[4]	W57
DDR4_DQS_T[5]	W51
DDR4_DQS_C[5]	W52
DDR4_DQS_T[6]	AC49
DDR4_DQS_C[6]	AB49
DDR4_DQS_T[7]	AC48

Pin Assignment

DDR4_DQS_C[7]	AB48
DDR4_ODT	N59
DDR4_PAR	R63
DDR4_RESET_N	AC57
DDR4_TEN	AC51

4.2 Pin Assignment for CLK

Net Name	Pin Name	Frequency
250MHZ_CLK_P	P55	250MHz
250MHZ_CLK_N	P56	250MHz
MGT_CLK1_P	BB13	156.25MHZ
MGT_CLK1_N	BB12	156.25MHZ
MGT_CLK2_P	AU15	156.25MHZ
MGT_CLK2_N	AU14	156.25MHZ
MGT_CLK3_P	AG15	156.25MHZ
MGT_CLK3_N	AG14	156.25MHZ
MGT_CLK4_P	AA15	156.25MHZ
MGT_CLK4_N	AA14	156.25MHZ
MGT_CLK5_P	CB13	156.25MHZ
MGT_CLK5_N	CB12	156.25MHZ
MGT_CLK6_P	BM13	156.25MHZ
MGT_CLK6_N	BM12	156.25MHZ
MGT_CLK7_P	BD13	156.25MHZ
MGT_CLK7_N	BD12	156.25MHZ
MGT_CLK8_P	AY13	156.25MHZ
MGT_CLK8_N	AY12	156.25MHZ
MGT_CLK9_P	AN15	156.25MHZ
MGT_CLK9_N	AN14	156.25MHZ
MGT_CLK10_P	AC15	156.25MHZ
MGT_CLK10_N	AC14	156.25MHZ
OSC2	AM49	100MHz
USR_CLK1_P	AP60	10MHz
USR_CLK1_N	AP61	10MHz
USR_CLK2_P	AR60	10MHz
USR_CLK2_N	AT60	10MHz
USR_CLK3_P	BM39	10MHz
USR_CLK3_N	BN39	10MHz

Pin Assignment

USR_CLK4_P	BM38	10MHz
USR_CLK4_N	BN38	10MHz
USR_CLK5_P	P37	10MHz
USR_CLK5_N	N37	10MHz
PCIE_CLK0_P	BN15	100MHz
PCIE_CLK0_N	BN14	100MHz
PCIE_CLK1_P	BT13	100MHz
PCIE_CLK1_N	BT12	100MHz
PCIE_CLK2_P	BL15	100MHz
PCIE_CLK2_N	BL14	100MHz

4.3 Pin Assignment for FMC

Net Name	J31/J32 Pin Name	Pin Name
FMC_HA00_P_CC	J32: F4	AY53
FMC_HA01_P_CC	J32: E2	AW51
FMC_HA02_P	J32: K7	AY48
FMC_HA03_P	J32: J6	BB48
FMC_HA04_P	J32: F7	BC46
FMC_HA05_P	J32: E6	AY52
FMC_HA06_P	J32: K10	BA50
FMC_HA07_P	J32: J9	BA45
FMC_HA08_P	J32: F10	AW50
FMC_HA09_P	J32: E9	BA46
FMC_HA10_P	J32: K13	AW49
FMC_HA11_P	J32: J12	AW54
FMC_HA12_P	J32: F13	BC51
FMC_HA13_P	J32: E12	AY47
FMC_HA14_P	J32: J15	BA56
FMC_HA15_P	J32: F16	BC56
FMC_HA16_P	J32: E15	BA51
FMC_HA17_P_CC	J32: K16	AY54
FMC_HA18_P	J32: J18	BB53
FMC_HA19_P	J32: F19	BA55
FMC_HA20_P	J32: E18	BB54
FMC_HA21_P	J32: K19	AY57
FMC_HA22_P	J32: J21	AW56
FMC_HA23_P	J32: K22	BC57

Pin Assignment

FMC_LA00_P_CC	J32: G6	BE53
FMC_LA01_P_CC	J32: D8	BD51
FMC_LA02_P	J32: H7	BE57
FMC_LA03_P	J32: G9	BH56
FMC_LA04_P	J32: H10	BH57
FMC_LA05_P	J32: D11	BG55
FMC_LA06_P	J32: C10	BF55
FMC_LA07_P	J32: H13	BG53
FMC_LA08_P	J32: G12	BH52
FMC_LA09_P	J32: D14	BE54
FMC_LA10_P	J32: C14	BG54
FMC_LA11_P	J32: H16	BD49
FMC_LA12_P	J32: G15	BH51
FMC_LA13_P	J32: D17	BH49
FMC_LA14_P	J32: C18	BD50
FMC_LA15_P	J32: H19	BF51
FMC_LA16_P	J32: G18	BE52
FMC_LA17_P_CC	J32: D20	BD61
FMC_LA18_P_CC	J32: C22	BB60
FMC_LA19_P	J32: H22	AY59
FMC_LA20_P	J32: G21	AW61
FMC_LA21_P	J32: H25	BA60
FMC_LA22_P	J32: G24	AY62
FMC_LA23_P	J32: D23	BA62
FMC_LA24_P	J32: H28	BB58
FMC_LA25_P	J32: G27	BC58
FMC_LA26_P	J32: D26	AW59
FMC_LA27_P	J32: C26	BC61
FMC_LA28_	J32: H31	BE58
FMC_LA29_P	J32: G30	BF62
FMC_LA30_P	J32: H34	BE62
FMC_LA31_P	J32: G33	BG59
FMC_LA32_P	J32: H37	BD59
FMC_LA33_P	J32: G36	BB63
FMCP_HA00_P_CC	J31: F4	F18
FMCP_HA01_P_CC	J31: E2	J20
FMCP_HA02_P	J31: K7	L17
FMCP_HA03_P	J31: J6	J16
FMCP_HA04_P	J31: F7	L15
FMCP_HA05_P	J31: E6	E17

Pin Assignment

FMCP_HA06_P	J31: K10	F17
FMCP_HA07_P	J31: J9	K17
FMCP_HA08_P	J31: F10	H18
FMCP_HA09_P	J31: E9	K20
FMCP_HA10_P	J31: K13	G15
FMCP_HA11_P	J31: J12	D20
FMCP_HA12_P	J31: F13	H16
FMCP_HA13_P	J31: E12	L18
FMCP_HA14_P	J31: J15	B19
FMCP_HA15_P	J31: F16	B16
FMCP_HA16_P	J31: E15	G19
FMCP_HA17_P_CC	J31: K16	D16
FMCP_HA18_P	J31: J18	E20
FMCP_HA19_P	J31: F19	A18
FMCP_HA20_P	J31: E18	E15
FMCP_HA21_P	J31: K19	C19
FMCP_HA22_P	J31: J21	B20
FMCP_HA23_P	J31: K22	A16
FMCP_LA00_P_CC	J31: G6	R22
FMCP_LA01_P_CC	J31: D8	N22
FMCP_LA02_P	J31: H7	N24
FMCP_LA03_P	J31: G9	N27
FMCP_LA04_P	J31: H10	N28
FMCP_LA05_P	J31: D11	U28
FMCP_LA06_P	J31: C10	P27
FMCP_LA07_P	J31: H13	T27
FMCP_LA08_P	J31: G12	M21
FMCP_LA09_P	J31: D14	K28
FMCP_LA10_P	J31: C14	R28
FMCP_LA11_P	J31: H16	J22
FMCP_LA12_P	J31: G15	K25
FMCP_LA13_P	J31: D17	M25
FMCP_LA14_P	J31: C18	J26
FMCP_LA15_P	J31: H19	L28
FMCP_LA16_P	J31: G18D	L26
FMCP_LA17_P_CC	J31: D20	G23
FMCP_LA18_P_CC	J31: C22	D22
FMCP_LA19_P	J31: H22	H28
FMCP_LA20_P	J31: G21	F25
FMCP_LA21_P	J31: H25	H27

Pin Assignment

FMCP_LA22_P	J31: G24	G26
FMCP_LA23_P	J31: D23	C28
FMCP_LA24_P	J31: H28	D26
FMCP_LA25_P	J31: G27	A28
FMCP_LA26_P	J31: D26	B26
FMCP_LA27_P	J31: C26	A26
FMCP_LA28_P	J31: H31	H23
FMCP_LA29_P	J31: G30	C27
FMCP_LA30_P	J31: H34	F22
FMCP_LA31_P	J31: G33	H21
FMCP_LA32_P	J31: H37	F28
FMCP_LA33_P	J31: G36	E28

4.4 Pin Assignment for QSFP

Net Name	Pin Name
QSFP1_RX1_P_1	BC2
QSFP1_RX1_N_1	BC1
QSFP1_RX2_P_1	BB4
QSFP1_RX2_N_1	BB3
QSFP1_RX3_P_1	BA6
QSFP1_RX3_N_1	BA5
QSFP1_RX4_P_1	BA2
QSFP1_RX4_N_1	BA1
QSFP1_RX5_P_1	AY4
QSFP1_RX5_N_1	AY3
QSFP1_RX6_P_1	AW6
QSFP1_RX6_N_1	AW5
QSFP1_RX7_P_1	AW2
QSFP1_RX7_N_1	AW1
QSFP1_RX8_P_1	AV4
QSFP1_RX8_N_1	AV3
QSFP1_TX1_P_1	BC11
QSFP1_TX1_N_1	BC10
QSFP1_TX2_P_1	BB9
QSFP1_TX2_N_1	BB8

Pin Assignment

QSFP1_TX3_P_1	BA11
QSFP1_TX3_N_1	BA10
QSFP1_TX4_P_1	AY9
QSFP1_TX4_N_1	AY8
QSFP1_TX5_P_1	AW11
QSFP1_TX5_N_1	AW10
QSFP1_TX6_P_1	AV13
QSFP1_TX6_N_1	AV12
QSFP1_TX7_P_1	AV9
QSFP1_TX7_N_1	AV8
QSFP1_TX8_P_1	AU11
QSFP1_TX8_N_1	AU10
QSFP1_RX1_P_2	AU6
QSFP1_RX1_N_2	AU5
QSFP1_RX2_P_2	AU2
QSFP1_RX2_N_2	AU1
QSFP1_RX3_P_2	AT4
QSFP1_RX3_N_2	AT3
QSFP1_RX4_P_2	AR6
QSFP1_RX4_N_2	AR5
QSFP1_RX5_P_2	AR2
QSFP1_RX5_N_2	AR1
QSFP1_RX6_P_2	AP4
QSFP1_RX6_N_2	AP3
QSFP1_RX7_P_2	AN6
QSFP1_RX7_N_2	AN5
QSFP1_RX8_P_2	AN2
QSFP1_RX8_N_2	AN1
QSFP1_TX1_P_2	AT13
QSFP1_TX1_N_2	AT12
QSFP1_TX2_P_2	AT9
QSFP1_TX2_N_2	AT8
QSFP1_TX3_P_2	AR11
QSFP1_TX3_N_2	AR10
QSFP1_TX4_P_2	AP13
QSFP1_TX4_N_2	AP12

Pin Assignment

QSFP1_TX5_P_2	AP9
QSFP1_TX5_N_2	AP8
QSFP1_TX6_P_2	AN11
QSFP1_TX6_N_2	AN10
QSFP1_TX7_P_2	AM13
QSFP1_TX7_N_2	AM12
QSFP1_TX8_P_2	AM9
QSFP1_TX8_N_2	AM8
QSFP2_RX1_P_1	AM4
QSFP2_RX1_N_1	AM3
QSFP2_RX2_P_1	AL6
QSFP2_RX2_N_1	AL5
QSFP2_RX3_P_1	AL2
QSFP2_RX3_N_1	AL1
QSFP2_RX4_P_1	AK4
QSFP2_RX4_N_1	AK3
QSFP2_RX5_P_1	AJ6
QSFP2_RX5_N_1	AJ5
QSFP2_RX6_P_1	AJ2
QSFP2_RX6_N_1	AJ1
QSFP2_RX7_P_1	AH4
QSFP2_RX7_N_1	AH3
QSFP2_RX8_P_1	AG6
QSFP2_RX8_N_1	AG5
QSFP2_TX1_P_1	AL11
QSFP2_TX1_N_1	AL10
QSFP2_TX2_P_1	AK13
QSFP2_TX2_N_1	AK12
QSFP2_TX3_P_1	AK9
QSFP2_TX3_N_1	AK8
QSFP2_TX4_P_1	AJ11
QSFP2_TX4_N_1	AJ10
QSFP2_TX5_P_1	AH13
QSFP2_TX5_N_1	AH12
QSFP2_TX6_P_1	AH9
QSFP2_TX6_N_1	AH8

Pin Assignment

QSFP2_TX7_P_1	AG11
QSFP2_TX7_N_1	AG10
QSFP2_TX8_P_1	AF13
QSFP2_TX8_N_1	AF12
QSFP2_RX1_P_2	AG2
QSFP2_RX1_N_2	AG1
QSFP2_RX2_P_2	AF4
QSFP2_RX2_N_2	AF3
QSFP2_RX3_P_2	AE6
QSFP2_RX3_N_2	AE5
QSFP2_RX4_P_2	AE2
QSFP2_RX4_N_2	AE1
QSFP2_RX5_P_2	AD4
QSFP2_RX5_N_2	AD3
QSFP2_RX6_P_2	AC6
QSFP2_RX6_N_2	AC5
QSFP2_RX7_P_2	AC2
QSFP2_RX7_N_2	AC1
QSFP2_RX8_P_2	AB4
QSFP2_RX8_N_2	AB3
QSFP2_TX1_P_2	AF9
QSFP2_TX1_N_2	AF8
QSFP2_TX2_P_2	AE11
QSFP2_TX2_N_2	AE10
QSFP2_TX3_P_2	AD13
QSFP2_TX3_N_2	AD12
QSFP2_TX4_P_2	AD9
QSFP2_TX4_N_2	AD8
QSFP2_TX5_P_2	AC11
QSFP2_TX5_N_2	AC10
QSFP2_TX6_P_2	AB9
QSFP2_TX6_N_2	AB8
QSFP2_TX7_P_2	AA11
QSFP2_TX7_N_2	AA10
QSFP2_TX8_P_2	Y9
QSFP2_TX8_N_2	Y8

4.5 Pin Assignment for PCIe

Net Name	Pin Name
PCIE_CLK0_P	BN15
PCIE_CLK0_N	BN14
PCIE_CLK1_P	BT13
PCIE_CLK1_N	BT12
PCIE_CLK2_P	BL15
PCIE_CLK2_N	BL14
MGT_CLK5_P	CB13
MGT_CLK5_N	CB12
MGT_CLK6_P	BM13
MGT_CLK6_N	BM12
PCIE_RX0_P	BK4
PCIE_RX0_N	BK3
PCIE_RX1_P	BL2
PCIE_RX1_N	BL1
PCIE_RX2_P	BL6
PCIE_RX2_N	BL5
PCIE_RX3_P	BM4
PCIE_RX3_N	BM3
PCIE_RX4_P	BN2
PCIE_RX4_N	BN1
PCIE_RX5_P	BN6
PCIE_RX5_N	BN5
PCIE_RX6_P	BP4
PCIE_RX6_N	BP3
PCIE_RX7_P	BR2
PCIE_RX7_N	BR1
PCIE_RX8_P	BR6
PCIE_RX8_N	BR5
PCIE_RX9_P	BT4
PCIE_RX9_N	BT3
PCIE_RX10_P	BU2
PCIE_RX10_N	BU1
PCIE_RX11_P	BU6

Pin Assignment

PCIE_RX11_N	BU5
PCIE_RX12_P	BV4
PCIE_RX12_N	BV3
PCIE_RX13_P	BW2
PCIE_RX13_N	BW1
PCIE_RX14_P	BW6
PCIE_RX14_N	BW5
PCIE_RX15_P	BY4
PCIE_RX15_N	BY3
PCIE_TX0_P	BK9
PCIE_TX0_N	BK8
PCIE_TX1_P	BL11
PCIE_TX1_N	BL10
PCIE_TX2_P	BM9
PCIE_TX2_N	BM8
PCIE_TX3_P	BN11
PCIE_TX3_N	BN10
PCIE_TX4_P	BP9
PCIE_TX4_N	BP8
PCIE_TX5_P	BR11
PCIE_TX5_N	BR10
PCIE_TX6_P	BT9
PCIE_TX6_N	BT8
PCIE_TX7_P	BU11
PCIE_TX7_N	BU10
PCIE_TX8_P	BV9
PCIE_TX8_N	BV8
PCIE_TX9_P	BW11
PCIE_TX9_N	BW10
PCIE_TX10_P	BY9
PCIE_TX10_N	BY8
PCIE_TX11_P	CA7
PCIE_TX11_N	CA6
PCIE_TX12_P	CA11
PCIE_TX12_N	CA10
PCIE_TX13_P	CB9

Pin Assignment

PCIE_TX13_N	CB8
PCIE_TX14_P	CC7
PCIE_TX14_N	CC6
PCIE_TX15_P	CC11
PCIE_TX15_N	CC10

4.6 Pin Assignment for UART

Net Name	Pin Name (FPGA)	Pin Name (USB-UART)
USB_UART_TX	AD19	21
USB_UART_RX	AE19	20

4.7 Pin Assignment for PMOD

Net Name	Pin Name
PMOD0_0	BD19
PMOD0_1	BE19
PMOD0_2	BE18
PMOD0_3	BF17
PMOD0_4	BD17
PMOD0_5	BE17
PMOD0_6	BC18
PMOD0_7	BC17
PMOD1_0	U21
PMOD1_1	U20
PMOD1_2	V19
PMOD1_3	U19
PMOD1_4	V18
PMOD1_5	U18
PMOD1_6	T21
PMOD1_7	T20

4.8 Pin Assignment for ETH PHY

Net Name	Pin Name (FPGA)
ENET_CLKOUT	CA35
ENET_SGMII_CLK_P	CA36
ENET_SGMII_CLK_N	CB36
ENET_MDC	CB35
ENET_SGMII_IN_P	BY37
ENET_SGMII_IN_N	CA37

Pin Assignment

ENET_SGMII_OUT_P	BV36
ENET_SGMII_OUT_N	BW36
ENET_MDIO	BV37
ENET_GPIO_0	BW37
ENET_PWDN_B_I_INT_B_O	CC36
ENET_RESET_B	CC32